

# The Fourier, Laplace, and z-Transforms

|     |                                                                             |    |
|-----|-----------------------------------------------------------------------------|----|
| 2.1 | Introduction .....                                                          | 17 |
| 2.2 | Fundamentals of the Fourier, Laplace, and z-Transforms .....                | 17 |
|     | Laplace Transform • Rational Laplace Transforms • Irrational Transforms •   |    |
|     | Discrete-Time Fourier Transform • z-Transform • Rational z-Transforms       |    |
| 2.3 | Applications and Examples .....                                             | 25 |
|     | Spectrum of a Signal Having a Rational Laplace Transform • Numerical        |    |
|     | Computation of the Fourier Transform • Solution of Differential Equations • |    |
|     | Solution of Difference Equations • Defining Terms                           |    |
|     | References .....                                                            | 31 |
|     | Further Reading .....                                                       | 31 |

Edward W. Kamen  
*School of Electrical and Computer Engineering,  
 Georgia Institute of Technology, Atlanta, GA*

## 2.1 Introduction

The study of signals and systems can be carried out in terms of either a time domain or a transform domain formulation. Both approaches are often used together in order to maximize our ability to deal with a particular problem arising in applications. This is very much the case in controls engineering where both time domain and transform domain techniques are extensively used in analysis and design. The transform domain approach to signals and systems is based on the transformation of functions using the Fourier, Laplace, and z-transforms. The fundamental aspects of these transforms are presented in this section along with some discussion on the application of these constructs.

The development in this chapter begins with the Fourier transform (FT), which can be viewed as a generalization of the Fourier series representation of a periodic function. The Fourier transform and Fourier series are named after Jean Baptiste Joseph Fourier (1768-1830), who first proposed in a 1807 paper that a series of sinusoidal harmonics could be used to represent the temperature distribution in a body. In 1822 Fourier wrote a book on his work, which was translated into English many years later (see [2]). It was also during the first part of the 1800s that Fourier was successful in constructing a frequency domain representation for aperiodic (nonperiodic) functions. This resulted in the FT, which provides a representation of a function  $f(t)$  of a real variable  $t$  in terms of the frequency components comprising the function. Much later (in the 1900s), a FT theory was developed for functions  $f(k)$  of an integer variable  $k$ . This resulted in the discrete-time Fourier transform (DTFT) and the N-point discrete Fourier transform (N-point DFT), both of which are briefly considered in this section.

Also during the early part of the 1800s, Pierre Simon Laplace (1749-1827) carried out his work on the generalization of the FT,

which resulted in the transform that now bears his name. The Laplace transform can be viewed as the FT with the addition of a real exponential factor to the integrand of the integral operation. This results in a transform that is a function of a complex variable  $s = \sigma + j\omega$ . Although the modification to the FT may not seem to be very major, in fact the Laplace transform is an extremely powerful tool in many application areas (such as controls) where the utility of the FT is somewhat limited. In this section, a brief presentation is given on the one-sided Laplace transform with much of the focus on rational transforms.

The discrete-time counterpart to the Laplace transform is the z-transform which was developed primarily during the 1950s (e.g., see [1], [3], and [5]). The one-sided z-transform is considered, along with the connection to the DTFT.

Applications and examples involving the Fourier, Laplace, and z-transforms are given in the second part of this section. There the presentation centers on the relationship between the pole locations of a rational transform and the frequency spectrum of the transformed function; the numerical computation of the FT; and the application of the Laplace and z-transforms to solving differential and difference equations. The application of the transforms to systems and controls is pursued in other chapters in this handbook.

## 2.2 Fundamentals of the Fourier, Laplace, and z-Transforms

Let  $f(t)$  be a real-valued function of the real-valued variable  $t$ ; that is, for any real number  $t$ ,  $f(t)$  is a real number. The function  $f(t)$  can be viewed as a signal that is a function of the continuous-time variable  $t$  (in units of seconds) and where  $t$  takes values from

$-\infty$  to  $\infty$ . The FT  $F(\omega)$  of  $f(t)$  is defined by

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt, \quad -\infty < \omega < \infty \quad (2.1)$$

where  $\omega$  is the frequency variable in radians per second (rad/s),  $j = \sqrt{-1}$  and  $e^{-j\omega t}$  is the complex exponential given by Euler's formula

$$e^{-j\omega t} = \cos(\omega t) - j \sin(\omega t) \quad (2.2)$$

Inserting Equation 2.2 into Equation 2.1 results in the following expression for the FT:

$$F(\omega) = R(\omega) + jI(\omega) \quad (2.3)$$

where  $R(\omega)$  and  $I(\omega)$  are the real and imaginary parts, respectively, of  $F(\omega)$  given by

$$\begin{aligned} R(\omega) &= \int_{-\infty}^{\infty} f(t) \cos(\omega t) dt \\ I(\omega) &= -\int_{-\infty}^{\infty} f(t) \sin(\omega t) dt \end{aligned} \quad (2.4)$$

From Equation 2.3, it is seen that in general the FT  $F(\omega)$  is a complex-valued function of the frequency variable  $\omega$ . For any value of  $\omega$ ,  $F(\omega)$  has a magnitude  $|F(\omega)|$  and an angle  $\angle F(\omega)$  given by

$$\begin{aligned} |F(\omega)| &= \sqrt{R^2(\omega) + I^2(\omega)} \\ \angle F(\omega) &= \tan^{-1} \left( \frac{I(\omega)}{R(\omega)} \right) \end{aligned} \quad (2.5)$$

where again  $R(\omega)$  and  $I(\omega)$  are the real and imaginary parts defined by Equation 2.4. The function  $|F(\omega)|$  represents the magnitude of the frequency components comprising  $f(t)$ , and thus the plot of  $|F(\omega)|$  vs.  $\omega$  is called the **magnitude spectrum** of  $f(t)$ . The function  $\angle F(\omega)$  represents the phase of the frequency components comprising  $f(t)$ , and thus the plot of  $\angle F(\omega)$  vs.  $\omega$  is called the **phase spectrum** of  $f(t)$ . Note that  $F(\omega)$  can be expressed in the polar form

$$F(\omega) = |F(\omega)| \exp[j\angle F(\omega)] \quad (2.6)$$

whereas the rectangular form of  $F(\omega)$  is given by Equation 2.3.

The function (or signal)  $f(t)$  is said to have a FT in the ordinary sense if the integral in Equation 2.1 exists for all real values of  $\omega$ . Sufficient conditions that ensure the existence of the integral are that  $f(t)$  have only a finite number of discontinuities, maxima, and minima over any finite interval of time and that  $f(t)$  be absolutely integrable. The latter condition means that

$$\int_{-\infty}^{\infty} |f(t)| dt < \infty \quad (2.7)$$

There are a number of functions  $f(t)$  of interest for which the integral in Equation 2.1 does not exist; for example, this is the case for the constant function  $f(t) = c$  for  $-\infty < t < \infty$ , where  $c$  is a nonzero real number. Since the integral in Equation 2.1 obviously does not exist in this case, the constant function

does not have a FT in the ordinary sense, but it does have a FT in the generalized sense, given by

$$\bar{F}(\omega) = 2\pi c \delta(\omega) \quad (2.8)$$

where  $\delta(\omega)$  is the impulse function. If Equation 2.8 is inserted into the inverse FT given by Equation 2.11, the result is the constant function  $f(t) = c$  for all  $t$ . This observation justifies taking Equation 2.8 as the definition of the (generalized) FT of the constant function.

The FT defined by Equation 2.1 can be viewed as an operator that maps a time function  $f(t)$  into a frequency function  $F(\omega)$ . This operation is often given by

$$F(\omega) = \mathfrak{F}[f(t)] \quad (2.9)$$

where  $\mathfrak{F}$  denotes the FT operator. From Equation 2.9, it is clear that  $f(t)$  can be recomputed from  $F(\omega)$  by applying the inverse FT operator denoted by  $\mathfrak{F}^{-1}$ ; that is,

$$f(t) = \mathfrak{F}^{-1}[F(\omega)] \quad (2.10)$$

The inverse operation is given by

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega \quad (2.11)$$

The FT satisfies a number of properties that are very useful in applications. These properties are listed in Table 2.1 in terms of functions  $f(t)$  and  $g(t)$ , whose transforms are  $F(\omega)$  and  $G(\omega)$ , respectively. Appearing in this table is the convolution  $f(t) * g(t)$  of  $f(t)$  and  $g(t)$ , defined by

$$f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau \quad (2.12)$$

Also in Table 2.1 is the convolution  $F(\omega) * G(\omega)$  given by

$$F(\omega) * G(\omega) = \int_{-\infty}^{\infty} F(\lambda) G(\omega - \lambda) d\lambda \quad (2.13)$$

From the properties in Table 2.1 and the generalized transform given by Equation 2.8, it is possible to determine the FT of many common functions. A list of FT of some common functions is given in Table 2.2.

## 2.2.1 Laplace Transform

Given the real-valued function  $f(t)$ , the **two-sided** (or **bilateral**) Laplace transform  $F(s)$  of  $f(t)$  is defined by

$$F(s) = \int_{-\infty}^{\infty} f(t) e^{-st} dt \quad (2.14)$$

where  $s$  is a complex variable. The **one-sided** (or **unilateral**) Laplace transform of  $f(t)$  is defined by

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt \quad (2.15)$$

Note that if  $f(t) = 0$  for all  $t < 0$ , the one-sided and two-sided Laplace transforms of  $f(t)$  are identical. In controls engineering, the one-sided Laplace transform is primarily used, and thus our presentation focuses on only the one-sided Laplace transform, which is referred to as the Laplace transform.

TABLE 2.1 Properties of the Fourier Transform.

| Property                              | Transform/Property                                                                                       |
|---------------------------------------|----------------------------------------------------------------------------------------------------------|
| Linearity                             | $\mathfrak{F}[af(t) + bg(t)] = aF(\omega) + bG(\omega)$ for any scalars $a, b$                           |
| Right or left shift in $t$            | $\mathfrak{F}[f(t - t_0)] = F(\omega) \exp(-j\omega t_0)$ for any $t_0$                                  |
| Time scaling                          | $\mathfrak{F}[f(at)] = (1/a)F(\omega/a)$ for any real number $a > 0$                                     |
| Time reversal                         | $\mathfrak{F}[f(-t)] = F(-\omega) = \overline{F(\omega)}$ = complex conjugate of $F(\omega)$             |
| Multiplication by a power of $t$      | $\mathfrak{F}[t^n f(t)] = j^n \frac{d^n}{d\omega^n} F(\omega)$ , $n = 1, 2, \dots$                       |
| Multiplication by $\exp(j\omega_0 t)$ | $\mathfrak{F}[f(t) \exp(j\omega_0 t)] = F(\omega - \omega_0)$ for any real number $\omega_0$             |
| Multiplication by $\sin(\omega_0 t)$  | $\mathfrak{F}[f(t) \sin(\omega_0 t)] = (j/2)[F(\omega + \omega_0) - F(\omega - \omega_0)]$               |
| Multiplication by $\cos(\omega_0 t)$  | $\mathfrak{F}[f(t) \cos(\omega_0 t)] = (1/2)[F(\omega + \omega_0) + F(\omega - \omega_0)]$               |
| Differentiation in the time domain    | $\mathfrak{F}\left[\frac{d^n}{dt^n} f(t)\right] = (j\omega)^n F(\omega)$ , $n = 1, 2, \dots$             |
| Multiplication in the time domain     | $\mathfrak{F}[f(t)g(t)] = \frac{1}{2\pi} [F(\omega) * G(\omega)]$                                        |
| Convolution in the time domain        | $\mathfrak{F}[f(t) * g(t)] = F(\omega)G(\omega)$                                                         |
| Duality                               | $\mathfrak{F}[F(t)] = 2\pi f(-\omega)$                                                                   |
| Parseval's theorem                    | $\int_{-\infty}^{\infty} f(t)g(t)dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(-\omega)G(\omega)d\omega$ |
| Special case of Parseval's theorem    | $\int_{-\infty}^{\infty} f^2(t)dt = \frac{1}{2\pi} \int_{-\infty}^{\infty}  F(\omega) ^2 d\omega$        |

TABLE 2.2 Common Fourier Transforms.

| $f(t)$                                                                                   | $F(\omega)$                                                                           |
|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| $\delta(t)$ = unit impulse                                                               | 1                                                                                     |
| $c$ , $-\infty < t < \infty$                                                             | $2\pi c\delta(\omega)$                                                                |
| $f(t) = \begin{cases} 1, & -T/2 \leq t \leq T/2 \\ 0, & \text{all other } t \end{cases}$ | $\frac{2}{\omega} \sin\left(\frac{T\omega}{2}\right)$                                 |
| $\frac{\sin(at)}{t}$                                                                     | $\begin{cases} \pi, & -a < \omega < a \\ 0, & \text{all other } \omega \end{cases}$   |
| $e^{-b t }$ , any $b > 0$                                                                | $\frac{2b}{\omega^2 + b^2}$                                                           |
| $e^{-bt^2}$ any $b > 0$                                                                  | $\sqrt{\frac{\pi}{b}} e^{-\omega^2/4b}$                                               |
| $e^{j\omega_0 t}$                                                                        | $2\pi\delta(\omega - \omega_0)$                                                       |
| $\cos(\omega_0 t + \theta)$                                                              | $\pi [e^{-j\theta}\delta(\omega + \omega_0) + e^{j\theta}\delta(\omega - \omega_0)]$  |
| $\sin(\omega_0 t + \theta)$                                                              | $j\pi [e^{-j\theta}\delta(\omega + \omega_0) - e^{j\theta}\delta(\omega - \omega_0)]$ |

Given a function  $f(t)$ , the set of all complex numbers  $s$  such that the integral in Equation 2.15 exists is called the **region of convergence** of the Laplace transform of  $f(t)$ . For example, if  $f(t)$  is the unit-step function  $u(t)$  given by  $u(t) = 1$  for  $t \geq 0$  and  $u(t) = 0$  for  $t < 0$ , the integral in Equation 2.15 exists for any  $s = \sigma + j\omega$  with real part  $\sigma > 0$ . Hence, the region of convergence is the set of all complex numbers  $s$  with positive real part, and, for any such  $s$ , the transform of the unit-step function  $u(t)$  is equal to  $1/s$ .

Given a function  $f(t)$ , if the region of convergence of the Laplace transform  $F(s)$  includes all complex numbers  $s = j\omega$  for  $\omega$  ranging from  $-\infty$  to  $\infty$ , then  $F(j\omega) = F(s)|_{s=j\omega}$  is well defined (i.e., exists) and is given by

$$F(j\omega) = \int_0^{\infty} f(t)e^{-j\omega t} dt \quad (2.16)$$

Then if  $f(t) = 0$  for  $t < 0$ , the right-hand side of Equation 2.16 is equal to the FT  $F(\omega)$  of  $f(t)$  (see Equation 2.1). Hence, the FT of  $f(t)$  is given by

$$F(\omega) = F(s)|_{s=j\omega} \quad (2.17)$$

(Note that we are denoting  $F(j\omega)$  by  $F(\omega)$ .) This fundamental result shows that the FT of  $f(t)$  can be computed directly from the Laplace transform  $F(s)$  if  $f(t) = 0$  for  $t < 0$  and the region

of convergence includes the imaginary axis of the complex plane (all complex numbers equal to  $j\omega$ ).

The Laplace transform defined by Equation 2.15 can be viewed as an operator, denoted by  $F(s) = L[f(t)]$  that maps a time function  $f(t)$  into the function  $F(s)$  of the complex variable  $s$ . The inverse Laplace transform operator is often denoted by  $L^{-1}$ , and is given by

$$f(t) = L^{-1}[F(s)] = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(s)e^{st} ds \quad (2.18)$$

The integral in Equation 2.18 is evaluated along the path  $s = c + j\omega$  in the complex plane from  $c - j\infty$  to  $c + j\infty$ , where  $c$  is any real number for which the path  $c + j\omega$  lies in the region of convergence of the transform  $F(s)$ . It is often possible to determine  $f(t)$  without having to use Equation 2.18; for example, this is the case when  $F(s)$  is a rational function of  $s$ . The computation of the Laplace transform or the inverse transform is often facilitated by using the properties of the Laplace transform, which are listed in Table 2.3. In this table,  $f(t)$  and  $g(t)$  are two functions with Laplace transforms  $F(s)$  and  $G(s)$ , respectively, and  $u(t)$  is the unit-step function defined by  $u(t) = 1$  for  $t \geq 0$  and  $u(t) = 0$  for  $t < 0$ . Using the properties in Table 2.3, it is possible to determine the Laplace transform of many common functions without having to use Equation 2.15. A list of common Laplace transforms is given in Table 2.4.

TABLE 2.3 Properties of the (One-Sided) Laplace Transform.

| Property                             | Transform/Property                                                                                                                  |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Linearity                            | $L[af(t) + bg(t)] = aF(s) + bG(s)$ for any scalars $a, b$                                                                           |
| Right shift in $t$                   | $L[f(t - t_0)u(t - t_0)] = F(s) \exp(-st_0)$ for any $t_0 > 0$                                                                      |
| Time scaling                         | $L[f(at)] = (1/a)F(s/a)$ for any real number $a > 0$                                                                                |
| Multiplication by a power of $t$     | $L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} F(s), n = 1, 2, \dots$                                                                       |
| Multiplication by $e^{\alpha t}$     | $L[f(t)e^{\alpha t}] = F(s - \alpha)$ for any real or complex number $\alpha$                                                       |
| Multiplication by $\sin(\omega_0 t)$ | $L[f(t) \sin(\omega_0 t)] = (j/2)[F(s + j\omega_0) - F(s - j\omega_0)]$                                                             |
| Multiplication by $\cos(\omega_0 t)$ | $L[f(t) \cos(\omega_0 t)] = (1/2)[F(s + j\omega_0) + F(s - j\omega_0)]$                                                             |
| Differentiation in the time domain   | $L\left[\frac{d}{dt} f(t)\right] = sF(s) - f(0)$                                                                                    |
| Second derivative                    | $L\left[\frac{d^2}{dt^2} f(t)\right] = s^2 F(s) - sf(0) - \frac{d}{dt} f(0)$                                                        |
| $n$ th derivative                    | $L\left[\frac{d^n}{dt^n} f(t)\right] = s^n F(s) - s^{n-1} f(0) - s^{n-2} \frac{d}{dt} f(0) - \dots - \frac{d^{n-1}}{dt^{n-1}} f(0)$ |
| Integration                          | $L\left[\int_0^t f(\tau) d\tau\right] = \frac{1}{s} F(s)$                                                                           |
| Convolution in the time domain       | $L[f(t) * g(t)] = F(s)G(s)$                                                                                                         |
| Initial-value theorem                | $f(0) = \lim_{s \rightarrow \infty} sF(s)$                                                                                          |
| Final-value theorem                  | If $f(t)$ has a finite limit $f(\infty)$ as $t \rightarrow \infty$ , then $f(\infty) = \lim_{s \rightarrow 0} sF(s)$                |

TABLE 2.4 Common Laplace Transforms.

| $f(t)$                              | Laplace Transform $F(s)$                            |
|-------------------------------------|-----------------------------------------------------|
| $u(t) = \text{unit-step function}$  | $\frac{1}{s}$                                       |
| $u(t) - u(t - T)$ for any $T > 0$   | $\frac{1 - e^{-Ts}}{s}$                             |
| $\delta(t) = \text{unit impulse}$   | 1                                                   |
| $\delta(t - t_0)$ for any $t_0 > 0$ | $e^{-st_0}$                                         |
| $t^n, t \geq 0$                     | $\frac{n!}{s^{n+1}}, n = 1, 2, \dots$               |
| $e^{-at}$                           | $\frac{1}{s+a}$                                     |
| $t^n e^{-at}$                       | $\frac{n!}{(s+a)^{n+1}}, n = 1, 2, \dots$           |
| $\cos(\omega t)$                    | $\frac{s}{s^2 + \omega^2}$                          |
| $\sin(\omega t)$                    | $\frac{\omega}{s^2 + \omega^2}$                     |
| $\cos^2 \omega t$                   | $\frac{s^2 + 2\omega^2}{s(s^2 + 4\omega^2)}$        |
| $\sin^2 \omega t$                   | $\frac{2\omega^2}{s(s^2 + 4\omega^2)}$              |
| $\sinh(at)$                         | $\frac{a}{s^2 - a^2}$                               |
| $\cosh(at)$                         | $\frac{s}{s^2 - a^2}$                               |
| $e^{-at} \cos(\omega t)$            | $\frac{s+a}{(s+a)^2 + \omega^2}$                    |
| $e^{-at} \sin(\omega t)$            | $\frac{\omega}{(s+a)^2 + \omega^2}$                 |
| $t \cos(\omega t)$                  | $\frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}$         |
| $t \sin(\omega t)$                  | $\frac{2\omega s}{(s^2 + \omega^2)^2}$              |
| $t e^{-at} \cos(\omega t)$          | $\frac{(s+a)^2 - \omega^2}{[(s+a)^2 + \omega^2]^2}$ |
| $t e^{-at} \sin(\omega t)$          | $\frac{2\omega(s+a)}{[(s+a)^2 + \omega^2]^2}$       |

## 2.2.2 Rational Laplace Transforms

The Laplace transform  $F(s)$  of a function  $f(t)$  is said to be a **rational function** of  $s$  if it can be written as a ratio of polynomials in  $s$ ; that is,

$$F(s) = \frac{N(s)}{D(s)} \quad (2.19)$$

where  $N(s)$  and  $D(s)$  are polynomials in the complex variable  $s$  given by

$$N(s) = b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0 \quad (2.20)$$

$$D(s) = s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0 \quad (2.21)$$

In Equations 2.20 and 2.21,  $m$  and  $n$  are positive integers and the coefficients  $b_m, b_{m-1}, \dots, b_1, b_0$  and  $a_{n-1}, \dots, a_1, a_0$  are real numbers. In Equation 2.19, it is assumed that  $N(s)$  and  $D(s)$  do not have any common factors. If there are common factors, they should always be cancelled. Also note that the polynomial  $D(s)$  is monic; that is, the coefficient of  $s^n$  is equal to 1. A rational function  $F(s)$  can always be written with a monic denominator polynomial  $D(s)$ . The integer  $n$ , which is the degree of  $D(s)$ , is called the order of the rational function  $F(s)$ . It is assumed that  $n \geq m$ , in which case  $F(s)$  is said to be a **proper rational function**. If  $n > m$ ,  $F(s)$  is said to be **strictly proper**.

Given a rational transform  $F(s) = N(s)/D(s)$  with  $N(s)$  and  $D(s)$  defined by Equations 2.20 and 2.21, let  $z_1, z_2, \dots, z_m$  denote the roots of the polynomial  $N(s)$ , and let  $p_1, p_2, \dots, p_n$  denote the roots of  $D(s)$ ; that is,  $N(z_i) = 0$  for  $i = 1, 2, \dots, m$  and  $D(p_i) = 0$  for  $i = 1, 2, \dots, n$ . In general, the  $z_i$  and the  $p_i$  may be real or complex numbers, but if any are complex, they must appear in complex conjugate pairs. The numbers  $z_1, z_2, \dots, z_m$  are called the **zeros** of the rational function  $F(s)$  since  $F(s) = 0$  when  $s = z_i$  for  $i = 1, 2, \dots, m$ ; and the numbers  $p_1, p_2, \dots, p_n$  are called the **poles** of  $F(s)$  since the magnitude  $|F(s)|$  becomes infinite as  $s$  approaches  $p_i$  for  $i = 1, 2, \dots, n$ .

If  $F(s)$  is strictly proper ( $n > m$ ) and the poles  $p_1, p_2, \dots, p_n$  of  $F(s)$  are distinct (nonrepeated), then  $F(s)$  has the partial fraction expansion

$$F(s) = \frac{c_1}{s - p_1} + \frac{c_2}{s - p_2} + \dots + \frac{c_n}{s - p_n} \quad (2.22)$$

where the  $c_i$  are the **residues** given by

$$c_i = [(s - p_i)F(s)]_{s=p_i}, \quad i = 1, 2, \dots, n \quad (2.23)$$

For a given value of  $i$ , the residue  $c_i$  is real if and only if the corresponding pole  $p_i$  is real, and  $c_i$  is complex if and only if  $p_i$  is complex.

From Equation 2.22 we see that the inverse Laplace transform  $f(t)$  is given by the following sum of exponential functions:

$$f(t) = c_1 e^{p_1 t} + c_2 e^{p_2 t} + \cdots + c_n e^{p_n t} \quad (2.24)$$

If all the poles  $p_1, p_2, \dots, p_n$  of  $F(s)$  are real numbers, then  $f(t)$  is a sum of real exponentials given by Equation 2.24. If  $F(s)$  has a pair of complex poles  $p = \sigma \pm j\omega$ , then  $f(t)$  contains the term

$$c e^{(\sigma + j\omega)t} + \bar{c} e^{(\sigma - j\omega)t} \quad (2.25)$$

where  $\bar{c}$  is the complex conjugate of  $c$ . Then writing  $c$  in the polar form  $c = |c|e^{j\theta}$ , we have

$$\begin{aligned} c e^{(\sigma + j\omega)t} + \bar{c} e^{(\sigma - j\omega)t} &= |c|e^{j\theta} e^{(\sigma + j\omega)t} + |c|e^{-j\theta} e^{(\sigma - j\omega)t} \\ &= |c|e^{\sigma t} \left[ e^{j(\omega t + \theta)} + e^{-j(\omega t + \theta)} \right] \end{aligned} \quad (2.26)$$

Finally, using Euler's formula, Equation 2.26 can be written in the form

$$c e^{(\sigma + j\omega)t} + \bar{c} e^{(\sigma - j\omega)t} = 2|c|e^{\sigma t} \cos(\omega t + \theta) \quad (2.27)$$

From Equation 2.27 it is seen that if  $F(s)$  has a pair of complex poles, then  $f(t)$  contains a sinusoidal term with an exponential amplitude factor  $e^{\sigma t}$ . Note that if  $\sigma = 0$  (so that the poles are purely imaginary), Equation 2.27 is purely sinusoidal.

If one of the poles (say  $p_1$ ) is repeated  $r$  times and the other  $n - r$  poles are distinct,  $F(s)$  has the partial fraction expansion

$$\begin{aligned} F(s) &= \frac{c_1}{s - p_1} + \frac{c_2}{(s - p_1)^2} + \cdots + \frac{c_r}{(s - p_1)^r} \\ &\quad + \frac{c_{r+1}}{s - p_{r+1}} + \cdots + \frac{c_n}{s - p_n} \end{aligned} \quad (2.28)$$

In Equation 2.28, the residues  $c_{r+1}, c_{r+2}, \dots, c_n$  are calculated as in the distinct-pole case; that is,

$$c_i = [(s - p_i)F(s)]_{s=p_i}, \quad i = r+1, r+2, \dots, n \quad (2.29)$$

and the residues  $c_1, c_2, \dots, c_r$  are given by

$$c_i = \frac{1}{(r-i)!} \left\{ \frac{d^{r-i}}{ds^{r-i}} [(s - p_1)^r F(s)] \right\}_{s=p_1} \quad (2.30)$$

Then, taking the inverse transform of Equation 2.28 yields

$$\begin{aligned} f(t) &= c_1 e^{p_1 t} + c_2 t e^{p_1 t} + \cdots + \frac{c_r}{(r-1)!} t^{r-1} e^{p_1 t} \\ &\quad + c_{r+1} e^{p_{r+1} t} + \cdots + c_n e^{p_n t} \end{aligned} \quad (2.31)$$

The above results reveal that the analytical form of the function  $f(t)$  depends directly on the poles of  $F(s)$ . In particular, if  $F(s)$  has a nonrepeated real pole  $p$ , then  $f(t)$  contains a real exponential term of the form  $c e^{pt}$  for some real constant  $c$ . If a real pole  $p$  is repeated  $r$  times, then  $f(t)$  contains terms of the form  $c_1 e^{pt}, c_2 t e^{pt}, \dots, c_r t^{r-1} e^{pt}$  for some real constants  $c_1, c_2, \dots, c_r$ . If  $F(s)$  has a nonrepeated complex pair  $\sigma \pm j\omega$

of poles, then  $f(t)$  contains a term of the form  $c e^{\sigma t} \cos(\omega t + \theta)$  for some real constants  $c$  and  $\theta$ . If the complex pair  $\sigma \pm j\omega$  is repeated  $r$  times,  $f(t)$  contains terms of the form  $c_1 e^{\sigma t} \cos(\omega t + \theta_1), c_2 t e^{\sigma t} \cos(\omega t + \theta_2), \dots, c_r t^{r-1} e^{\sigma t} \cos(\omega t + \theta_r)$  for some real constants  $c_1, c_2, \dots, c_r$  and  $\theta_1, \theta_2, \dots, \theta_r$ . These results are summarized in Table 2.5.

If  $F(s)$  is proper, but not strictly proper (so that  $n = m$  in Equations 2.20 and 2.21), then using long division  $F(s)$  can be written in the form

$$F(s) = b_n + \frac{R(s)}{D(s)} \quad (2.32)$$

where the degree of  $R(s)$  is strictly less than  $n$ . Then  $R(s)/D(s)$  can be expanded via partial fractions as was done in the case when  $F(s)$  is strictly proper. Note that for  $F(s)$  given by Equation 2.32, the inverse Laplace transform  $f(t)$  contains the impulse  $b_n \delta(t)$ . Hence, having  $n = m$  in  $F(s)$  results in an impulsive term in the inverse transform.

From the relationship between the poles of  $F(s)$  and the analytical form of  $f(t)$ , it follows that  $f(t)$  converges to zero as  $t \rightarrow \infty$  if and only if all the poles  $p_1, p_2, \dots, p_n$  of  $F(s)$  have real parts that are strictly less than zero; that is,  $\text{Re}(p_i) < 0$  for  $i = 1, 2, \dots, n$ . This condition is equivalent to requiring that all the poles be located in the **open left half-plane (OLHP)**, which is the region in the complex plane to the left of the imaginary axis.

It also follows from the relationship between the poles of  $F(s)$  and the form of  $f(t)$  that  $f(t)$  has a finite limit  $f(\infty)$  as  $t \rightarrow \infty$  if and only if all the poles of  $F(s)$  have real parts that are less than zero, except that  $F(s)$  may have a nonrepeated pole at  $s = 0$ . In mathematical terms, the conditions for the existence of a finite limit  $f(\infty)$  are

$$\text{Re}(p_i) < 0 \quad \text{for all poles } p_i \neq 0 \quad (2.33)$$

$$\begin{aligned} \text{If } p_i = 0 &\quad \text{is a pole of } F(s), \\ &\quad \text{then } p_i \text{ is nonrepeated} \end{aligned} \quad (2.34)$$

If the conditions in Equations 2.33 and 2.34 are satisfied, the limiting value  $f(\infty)$  is given by

$$f(\infty) = [sF(s)]_{s=0} \quad (2.35)$$

The relationship in Equation 2.35 is a restatement of the final-value theorem (given in Table 2.3) in the case when  $F(s)$  is rational and the poles of  $F(s)$  satisfy the conditions in Equations 2.33 and 2.34.

### 2.2.3 Irrational Transforms

The Laplace transform  $F(s)$  of a function  $f(t)$  is said to be an **irrational function** of  $s$  if it is not rational; that is,  $F(s)$  cannot be expressed as a ratio of polynomials in  $s$ . For example,  $F(s) = e^{-t_0 s}/s$  is irrational since the exponential function  $e^{-t_0 s}$  cannot be expressed as a ratio of polynomials in  $s$ . In this case, the inverse transform  $f(t)$  is equal to  $u(t - t_0)$  where  $u(t)$  is the unit-step function.

Given any function  $f(t)$  with transform  $F(s)$  and given any real number  $t_0 > 0$ , the transform of the time-shifted (or time-

TABLE 2.5 Relationship Between the Poles of  $F(s)$  and the Form of  $f(t)$ .

| Pole Locations of $F(s)$                                    | Corresponding Terms in $f(t)$                                     |
|-------------------------------------------------------------|-------------------------------------------------------------------|
| Nonrepeated real pole at $s = p$                            | $ce^{pt}$                                                         |
| Real pole at $s = p$ repeated $r$ times                     | $\sum_{i=1}^r c_i t^{i-1} e^{pt}$                                 |
| Nonrepeated complex pair at $s = \sigma \pm j\omega$        | $ce^{\sigma t} \cos(\omega t + \theta)$                           |
| Complex pair at $s = \sigma \pm j\omega$ repeated $r$ times | $\sum_{i=1}^r c_i t^{i-1} e^{\sigma t} \cos(\omega t + \theta_i)$ |

delayed) function  $f(t-t_0)u(t-t_0)$  is equal to  $F(s)e^{-t_0 s}$ . Time-delayed signals arise in systems with time delays, and thus irrational transforms appear in the study of systems with time delays. Also, any function  $f(t)$  that is of finite duration in time has a transform  $F(s)$  that is irrational. For instance, suppose that

$$f(t) = \gamma(t) [u(t-t_0) - u(t-t_1)], \quad 0 \leq t_0 < t_1 \quad (2.36)$$

so that  $f(t) = \gamma(t)$  for  $t_0 \leq t < t_1$ , and  $f(t) = 0$  for all other  $t$ . Then  $f(t)$  can be written in the form

$$f(t) = \gamma_0(t-t_0)u(t-t_0) - \gamma_1(t-t_1)u(t-t_1) \quad (2.37)$$

where  $\gamma_0(t) = \gamma(t+t_0)$  and  $\gamma_1(t) = \gamma(t+t_1)$ . Taking the Laplace transform of Equation 2.37 yields

$$F(s) = \Gamma_0(s)e^{-t_0 s} - \Gamma_1(s)e^{-t_1 s} \quad (2.38)$$

where  $\Gamma_0(s)$  and  $\Gamma_1(s)$  are the transforms of  $\gamma_0(t)$  and  $\gamma_1(t)$ , respectively. Note that by Equation 2.38, the transform  $F(s)$  is an irrational function of  $s$ .

To illustrate the above constructions, suppose that

$$f(t) = e^{-at} [u(t-1) - u(t-2)] \quad (2.39)$$

Writing  $f(t)$  in the form of Equation 2.37 gives

$$f(t) = e^{-a} e^{-a(t-1)} u(t-1) - e^{-2a} e^{-a(t-2)} u(t-2) \quad (2.40)$$

Then, transforming Equation 2.40 yields

$$F(s) = [e^{-(s+a)} - e^{-2(s+a)}] \frac{1}{s+a} \quad (2.41)$$

Clearly,  $F(s)$  is an irrational function of  $s$ .

## 2.2.4 Discrete-Time Fourier Transform

Let  $f(k)$  be a real-valued function of the integer-valued variable  $k$ . The function  $f(k)$  can be viewed as a discrete-time signal; in particular,  $f(k)$  may be a sampled version of a continuous-time signal  $f(t)$ . More precisely,  $f(k)$  may be equal to the sample values  $f(kT)$  of a signal  $f(t)$  with  $t$  evaluated at the sample times  $t = kT$ , where  $T$  is the sampling interval. In mathematical terms, the sampled signal is given by

$$f(k) = f(t)|_{t=kT} = f(kT), \quad k = 0, \pm 1, \pm 2, \dots \quad (2.42)$$

Note that we are denoting  $f(kT)$  by  $f(k)$ . The FT of a function  $f(k)$  of an integer variable  $k$  is defined by

$$F(\Omega) = \sum_{k=-\infty}^{\infty} f(k) e^{-j\Omega k}, \quad -\infty < \Omega < \infty \quad (2.43)$$

where  $\Omega$  is interpreted as the real frequency variable. The transform  $F(\Omega)$  is called the discrete-time Fourier transform (DTFT) since it can be viewed as the discrete-time counterpart of the FT defined above. The DTFT is directly analogous to the FT, so that all the properties of the FT discussed above carry over to the DTFT. In particular, as is the case for the FT, the DTFT  $F(\Omega)$  is in general a complex-valued function of the frequency variable  $\Omega$ , and thus  $F(\Omega)$  must be specified in terms of a magnitude function  $|F(\Omega)|$  and an angle function  $\angle F(\Omega)$ . The magnitude function  $|F(\Omega)|$  (respectively, the angle function  $\angle F(\Omega)$ ) displays the magnitude (respectively, the phase) of the frequency components comprising  $f(k)$ . All of the properties of the FT listed in Table 2.1 have a counterpart for the DTFT, but this will not be pursued here.

In contrast to the FT, the DTFT  $F(\Omega)$  is always a periodic function of the frequency variable  $\Omega$  with period  $2\pi$ ; that is,

$$F(\Omega + 2\pi) = F(\Omega) \quad \text{for } -\infty < \Omega < \infty \quad (2.44)$$

As a result of the periodicity property in Equation 2.44, it is necessary to specify  $F(\Omega)$  over a  $2\pi$  interval only, such as 0 to  $2\pi$  or  $-\pi$  to  $\pi$ . Given  $F(\Omega)$  over any  $2\pi$  interval,  $f(k)$  can be recomputed using the inverse DTFT. In particular, if  $F(\Omega)$  is specified over the interval  $-\pi < \Omega < \pi$ ,  $f(k)$  can be computed from the relationship

$$f(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(\Omega) e^{jk\Omega} d\Omega \quad (2.45)$$

In practice, the DTFT  $F(\Omega)$  is usually computed only for a discrete set of values of the frequency variable  $\Omega$ . This is accomplished by using the  $N$ -point discrete Fourier transform ( $N$ -point DFT) of  $f(k)$  given by

$$F_n = \sum_{k=0}^{N-1} f(k) e^{-j2\pi kn/N}, \quad n = 0, 1, \dots, N-1 \quad (2.46)$$

where  $N$  is a positive integer. If  $f(k) = 0$  for  $k < 0$  and  $k \geq N$ , comparing Equations 2.46 and 2.43 reveals that

$$F_n = F\left(\frac{2\pi n}{N}\right), \quad n = 0, 1, \dots, N-1 \quad (2.47)$$

Hence, the DFT  $F_n$  is equal to the values of the DTFT  $F(\Omega)$  with  $\Omega$  evaluated at the discrete points  $\Omega = 2\pi n/N$  for  $n = 0, 1, 2, \dots, N-1$ .

The computation of the DFT  $F_n$  given by Equation 2.46 can be carried out using a fast algorithm called the Fast Fourier transform (FFT). The inverse FFT can be used to compute  $f(k)$  from  $F_n$ . A development of the FFT is beyond the scope of this section (see "Further Reading").

### 2.2.5 z-Transform

Given the function  $f(k)$ , the **two-sided** (or **bilateral**) z-transform  $F(z)$  of  $f(k)$  is defined by

$$F(z) = \sum_{k=-\infty}^{\infty} f(k)z^{-k} \quad (2.48)$$

where  $z$  is a complex variable. The **one-sided** (or **unilateral**) z-transform of  $f(k)$  is defined by

$$F(z) = \sum_{k=0}^{\infty} f(k)z^{-k} \quad (2.49)$$

Note that if  $f(k) = 0$  for  $k = -1, -2, \dots$ , the one-sided and two-sided z-transforms of  $f(k)$  are the same. As is the case with the Laplace transform, in controls engineering the one-sided version is the most useful, and thus the development given below is restricted to the one-sided z-transform, which is referred to as the z-transform.

Given  $f(k)$ , the set of all complex numbers  $z$  such that the summation in Equation 2.49 exists is called the region of convergence of the z-transform of  $f(k)$ . If the region of convergence of the z-transform includes all complex numbers  $z = e^{j\Omega}$  for  $\Omega$  ranging from  $-\infty$  to  $\infty$ , then  $F(e^{j\Omega}) = F(z)|_{z=e^{j\Omega}}$  is well defined (i.e., exists) and is given by

$$F(e^{j\Omega}) = \sum_{k=0}^{\infty} f(k) (e^{j\Omega})^{-k} \quad (2.50)$$

But  $(e^{j\Omega})^{-k} = e^{-j\Omega k}$ , and thus Equation 2.50 can be rewritten as

$$F(e^{j\Omega}) = \sum_{k=0}^{\infty} f(k) e^{-j\Omega k} \quad (2.51)$$

Then if  $f(k) = 0$  for all  $k < 0$ , the right-hand side of Equation 2.51 is equal to the DTFT  $F(\Omega)$  of  $f(k)$  (see Equation 2.43). Therefore, the DTFT of  $f(k)$  is given by

$$F(\Omega) = F(z)|_{z=e^{j\Omega}} \quad (2.52)$$

This result shows that the DTFT of  $f(k)$  can be computed directly from the z-transform  $F(z)$  if  $f(k) = 0$  for all  $k < 0$  and the region of convergence includes all complex numbers  $z = e^{j\Omega}$  with  $-\infty < \Omega < \infty$ . Note that since  $|e^{j\Omega}| = 1$  for any value of  $\Omega$  and  $\angle e^{j\Omega} = \Omega$ , the set of complex numbers  $z = e^{j\Omega}$  comprises the unit circle of the complex plane. Hence, the DTFT of  $f(k)$  is equal to the values of the z-transform on the unit circle of the complex plane, assuming that the region of convergence of  $F(z)$  includes the unit circle.

The z-transform defined by Equation 2.49 can be viewed as an operator, denoted by  $F(z) = Z[f(k)]$ , that maps a discrete-time function  $f(k)$  into the function  $F(z)$  of the complex variable  $z$ . The inverse z-transform operation is denoted by  $f(k) = Z^{-1}[F(z)]$ . As discussed below, when  $F(z)$  is a rational function of  $z$ , the inverse transform can be computed using long division

or by carrying out a partial fraction expansion of  $F(z)$ . The computation of the z-transform or the inverse z-transform is often facilitated by using the properties of the z-transform given in Table 2.6. In this table,  $f(k)$  and  $g(k)$  are two functions with z-transforms  $F(z)$  and  $G(z)$ , respectively, and  $u(k)$  is the unit-step function defined by  $u(k) = 1$  for  $k \geq 0$  and  $u(k) = 0$  for  $k < 0$ . A list of common z-transforms is given in Table 2.7. In Table 2.7, the function  $\delta(k)$  is the unit pulse defined by  $\delta(0) = 1$ ,  $\delta(k) = 0$  for  $k \neq 0$ .

### 2.2.6 Rational z-Transforms

As is the case for the Laplace transform, the z-transform  $F(z)$  is often a rational function of  $z$ ; that is,  $F(z)$  is given by

$$F(z) = \frac{N(z)}{D(z)} \quad (2.53)$$

where  $N(z)$  and  $D(z)$  are polynomials in the complex variable  $z$  given by

$$N(z) = b_m z^m + b_{m-1} z^{m-1} + \dots + b_1 z + b_0 \quad (2.54)$$

$$D(z) = z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0 \quad (2.55)$$

It is assumed that the order  $n$  of  $F(z)$  is greater than or equal to  $m$ , and thus  $F(z)$  is proper. The poles and zeros of  $F(z)$  are defined in the same way as given above for rational Laplace transforms.

When the transform  $F(z)$  is in the rational form in Equation 2.53, the inverse z-transform  $f(k)$  can be computed by expanding  $F(z)$  into a power series in  $z^{-1}$  by dividing  $D(z)$  into  $N(z)$  using long division. The values of the function  $f(k)$  are then read off from the coefficients of the power series expansion. The first few steps of the process are carried out below:

$$\begin{array}{r} z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0 \overline{) b_m z^{m-n} + (b_{m-1} - a_{n-1} b_m) z^{m-n-1} + \dots} \\ \underline{b_m z^m + b_{m-1} z^{m-1} \phantom{+ \dots}} \phantom{+ \dots} \\ b_m z^m + a_{n-1} b_m z^{m-1} \phantom{+ \dots} \phantom{+ \dots} \\ \underline{(b_{m-1} - a_{n-1} b_m) z^{m-1} \phantom{+ \dots}} \phantom{+ \dots} \\ (b_{m-1} - a_{n-1} b_m) z^{m-1} \phantom{+ \dots} \phantom{+ \dots} \\ \underline{\phantom{(b_{m-1} - a_{n-1} b_m) z^{m-1} + \dots}} \phantom{+ \dots} \\ \vdots \end{array}$$

Since the value of  $f(k)$  is equal to the coefficient of  $z^{-k}$  in the power series expansion of  $F(z)$ , it follows from the above division process that  $f(n-m) = b_m$ ,  $f(n-m+1) = b_{m-1} - a_{n-1} b_m$ , and so on.

To express the inverse z-transform  $f(k)$  in closed form, it is necessary to expand  $F(z)$  via partial fractions. It turns out that the form of the inverse z-transform  $f(k)$  is simplified if  $F(z)/z = N(z)/D(z)z$  is expanded by partial fractions. Note that  $F(z)/z$  is strictly proper since  $F(z)$  is assumed to be proper.

Letting  $p_1, p_2, \dots, p_n$  denote the poles of  $F(z)$ , if the  $p_i$  are distinct and are nonzero, then  $F(z)/z$  has the partial fraction expansion

$$\frac{F(z)}{z} = \frac{c_0}{z} + \frac{c_1}{z-p_1} + \frac{c_2}{z-p_2} + \dots + \frac{c_n}{z-p_n} \quad (2.56)$$

TABLE 2.6 Properties of the (One-Sided) z-Transform.

| Property                           | Transform/Property                                                                                                       |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Linearity                          | $Z[af(k) + bg(k)] = aF(z) + bG(z)$ for any scalars $a, b$                                                                |
| Right shift of $f(k)u(k)$          | $Z[f(k-q)u(k-q)] = z^{-q}F(z)$ for any integer $q \geq 1$                                                                |
| Right shift of $f(k)$              | $Z[f(k-1)] = z^{-1}F(z) + f(-1)$                                                                                         |
|                                    | $Z[f(k-2)] = z^{-2}F(z) + f(-2) + z^{-1}f(-1)$                                                                           |
|                                    | $\vdots$                                                                                                                 |
| Left shift in time                 | $Z[f(k-q)] = z^{-q}F(z) + f(-q) + z^{-1}f(-q+1) + \cdots + z^{-q+1}f(-1)$                                                |
|                                    | $Z[f(k+1)] = zF(z) - f(0)z$                                                                                              |
|                                    | $Z[f(k+2)] = z^2F(z) - f(0)z^2 - f(1)z$                                                                                  |
|                                    | $\vdots$                                                                                                                 |
|                                    | $Z[f(k+q)] = z^qF(z) - f(0)z^q - f(1)z^{q-1} - \cdots - f(q-1)z$                                                         |
| Multiplication by $k$              | $Z[kf(k)] = -z \frac{d}{dz} F(z)$                                                                                        |
| Multiplication by $k^2$            | $Z[k^2f(k)] = z \frac{d}{dz} F(z) + z^2 \frac{d^2}{dz^2} F(z)$                                                           |
| Multiplication by $a^k$            | $Z[a^k f(k)] = F\left(\frac{z}{a}\right)$                                                                                |
| Multiplication by $\cos(\Omega k)$ | $Z[\cos(\Omega k)f(k)] = \frac{1}{2} [F(e^{j\Omega}z) + F(e^{-j\Omega}z)]$                                               |
| Multiplication by $\sin(\Omega k)$ | $Z[\sin(\Omega k)f(k)] = \frac{j}{2} [F(e^{j\Omega}z) - F(e^{-j\Omega}z)]$                                               |
| Summation                          | $\left[ \sum_{i=0}^k f(i) \right] = \frac{z}{z-1} F(z)$                                                                  |
| Convolution                        | $Z[f(k) * g(k)] = F(z)G(z)$                                                                                              |
| Initial-value theorem              | $f(0) = \lim_{z \rightarrow \infty} F(z)$                                                                                |
| Final-value theorem                | If $f(k)$ has a finite limit $f(\infty)$ as $k \rightarrow \infty$ , then $f(\infty) = \lim_{z \rightarrow 1} (z-1)F(z)$ |

where the residues are given by

$$c_0 = F(0) \quad (2.57)$$

$$c_i = \left[ (z - p_i) \frac{F(z)}{z} \right]_{z=p_i}, \quad i = 1, 2, \dots, n \quad (2.58)$$

Then multiplying both sides of Equation 2.56 by  $z$  gives

$$F(z) = c_0 + \frac{c_1 z}{z - p_1} + \frac{c_2 z}{z - p_2} + \cdots + \frac{c_n z}{z - p_n} \quad (2.59)$$

and taking the inverse z-transform gives

$$f(k) = c_0 \delta(k) + c_1 p_1^k + c_2 p_2^k + \cdots + c_n p_n^k \quad (2.60)$$

If the poles  $p_1, p_2, \dots, p_n$  of  $F(z)$  are real numbers, then from Equation 2.60 it is seen that  $f(k)$  is the sum of geometric functions of the form  $cp^k$ , plus a pulse function  $c_0 \delta(k)$  if  $c_0 \neq 0$ . If  $F(z)$  has a pair of complex poles given in polar form by  $\sigma e^{\pm j\Omega}$ , then it can be shown that  $f(k)$  contains a sinusoidal term of the form

$$c\sigma^k \cos(\Omega k + \theta) \quad (2.61)$$

for some constants  $c$  and  $\theta$ .

If  $F(z)$  has a real pole  $p$  that is repeated  $r$  times, then  $f(k)$  contains the terms  $c_1 p^k, c_2 k p^k, \dots, c_r k^{r-1} p^k$ ; and if  $F(z)$  has a pair of complex poles given by  $\sigma e^{\pm j\Omega}$  that is repeated  $r$  times, then  $f(k)$  contains terms of the form  $c_1 \sigma^k \cos(\Omega k + \theta_1), c_2 k \sigma^k \cos(\Omega k + \theta_2), \dots, c_r k^{r-1} \sigma^k \cos(\Omega k + \theta_r)$  for some constants  $c_1, c_2, \dots, c_r$  and  $\theta_1, \theta_2, \dots, \theta_r$ . These results are summarized in Table 2.8.

From the relationship between the poles of  $F(z)$  and the analytical form of  $f(k)$ , it follows that  $f(k)$  converges to zero as

TABLE 2.7 Common z-Transform Pairs.

| $f(k)$                           | z-Transform $F(z)$                                             |
|----------------------------------|----------------------------------------------------------------|
| $u(k)$ = unit-step function      | $\frac{z}{z-1}$                                                |
| $u(k) - u(k-N), N = 1, 2, \dots$ | $\frac{z^{N-1}}{z^{N-1}(z-1)}, N = 1, 2, \dots$                |
| $\delta(k)$ = unit pulse         | 1                                                              |
| $\delta(k-q), q = 1, 2, \dots$   | $\frac{1}{z^q}, q = 1, 2, \dots$                               |
| $a^k, a$ real or complex         | $\frac{z}{z-a}$                                                |
| $k$                              | $\frac{z}{(z-1)^2}$                                            |
| $k+1$                            | $\frac{z^2}{(z-1)^2}$                                          |
| $k^2$                            | $\frac{z(z+1)}{(z-1)^3}$                                       |
| $ka^k$                           | $\frac{az}{(z-a)^2}$                                           |
| $k^2 a^k$                        | $\frac{az(z+a)}{(z-a)^3}$                                      |
| $k(k+1)a^k$                      | $\frac{2az^2}{(z-a)^3}$                                        |
| $\cos(\Omega k)$                 | $\frac{z^2 - (\cos \Omega)z}{z^2 - (2 \cos \Omega)z + 1}$      |
| $\sin(\Omega k)$                 | $\frac{(\sin \Omega)z}{z^2 - (2 \cos \Omega)z + 1}$            |
| $a^k \cos(\Omega k)$             | $\frac{z^2 - (a \cos \Omega)z}{z^2 - (2a \cos \Omega)z + a^2}$ |
| $a^k \sin(\Omega k)$             | $\frac{(a \sin \Omega)z}{z^2 - (2a \cos \Omega)z + a^2}$       |

$k \rightarrow \infty$  if and only if all the poles  $p_1, p_2, \dots, p_n$  of  $F(z)$  have magnitudes that are strictly less than one; that is,  $|p_i| < 1$  for  $i = 1, 2, \dots, n$ . This is equivalent to requiring that all the poles be located on the open unit disk of the complex plane, which is the region of the complex plane consisting of all complex numbers whose magnitude is strictly less than one.

It also follows from the relationship between pole locations

$N(s)/D(s)$  where  $N(s)$  and  $D(s)$  are polynomials in  $s$  given by Equations 2.20 and 2.21. It was noted in the previous section that if the region of convergence of  $F(s)$  includes the imaginary axis ( $j\omega$ -axis) of the complex plane, then the FT  $F(\omega)$  is equal to the Laplace transform  $F(s)$  with  $s = j\omega$ . When  $F(s)$  is rational, it turns out that the region of convergence includes the  $j\omega$ -axis if and only if all the poles of  $F(s)$  lie in the OLHP; thus, in this case, the FT is given by

$$F(\omega) = F(s)|_{s=j\omega} \quad (2.69)$$

For example, if  $f(t) = ce^{-at}$  for  $t \geq 0$  with  $a > 0$ , then  $F(s) = c/(s+a)$  which has a single pole at  $s = -a$ . Since the point  $-a$  lies in the OLHP, the FT of the exponential function  $ce^{-at}$  is

$$F(\omega) = \frac{c}{j\omega + a} \quad (2.70)$$

It follows from Equation 2.70 that the 3-dB bandwidth is equal to  $a$ . Hence, the farther over in the OLHP the pole  $-a$  is (i.e., the larger  $a$  is), the larger the bandwidth of the signal. Since the rate of decay to zero of  $ce^{-at}$  increases as  $a$  is increased, this result again confirms the property that shorter duration time signals have wider bandwidths. In the case when  $c = a = 1$ , a plot of the magnitude spectrum  $|F(\omega)|$  is shown in Figure 2.2. For any real values of  $a$  and  $c$ , the magnitude spectrum rolls off to zero at the rate of 20 dB/decade where a decade is a factor of ten in frequency.

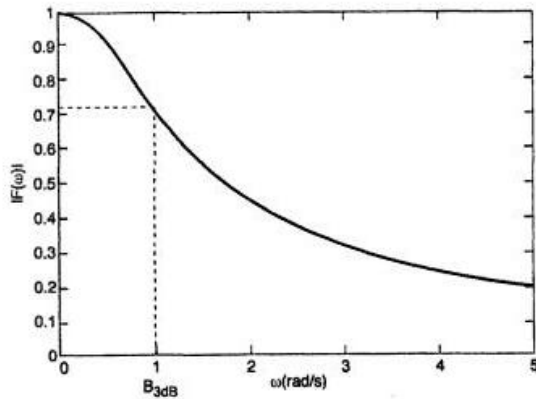


Figure 2.2 Magnitude spectrum of the exponential function  $e^{-t}$ .

As another example, consider the signal  $f(t)$  whose Laplace transform is

$$F(s) = \frac{c}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (2.71)$$

where  $\omega_n$  is assumed to be strictly positive ( $\omega_n > 0$ ). In this case,  $F(s)$  has two poles  $p_1$  and  $p_2$  given by

$$\begin{aligned} p_1 &= -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1} \\ p_2 &= -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1} \end{aligned} \quad (2.72)$$

When  $\zeta > 1$ , both poles are real, nonrepeated, and lie in the OLHP (assuming that  $\omega_n > 0$ ). As  $\zeta \rightarrow \infty$ , the pole  $p_1$  moves along the negative real axis to the origin of the complex plane and the pole  $p_2$  goes to  $-\infty$  along the negative axis of the complex plane. For  $\zeta > 1$ ,  $F(s)$  can be expanded by partial fractions as follows:

$$F(s) = \frac{c}{(s-p_1)(s-p_2)} = \frac{c}{p_1-p_2} \left[ \frac{1}{s-p_1} - \frac{1}{s-p_2} \right] \quad (2.73)$$

Taking the inverse Laplace transform gives

$$f(t) = \frac{c}{p_1-p_2} [e^{p_1 t} - e^{p_2 t}] \quad (2.74)$$

and thus  $f(t)$  is a sum of two decaying real exponentials. Since both poles lie in the OLHP, the FT  $F(\omega)$  is given by

$$F(\omega) = \frac{c}{\omega_n^2 - \omega^2 + j(2\zeta\omega_n\omega)} \quad (2.75)$$

For the case when  $c = \omega_n^2 = 100$  and  $\zeta = 2$ , the plot of the magnitude spectrum  $|F(\omega)|$  is given in Figure 2.3. In this case, the spectral content of the signal  $f(t)$  rolls off to zero at the rate of 40dB/decade, starting with the peak magnitude of 1 at  $\omega = 0$ .

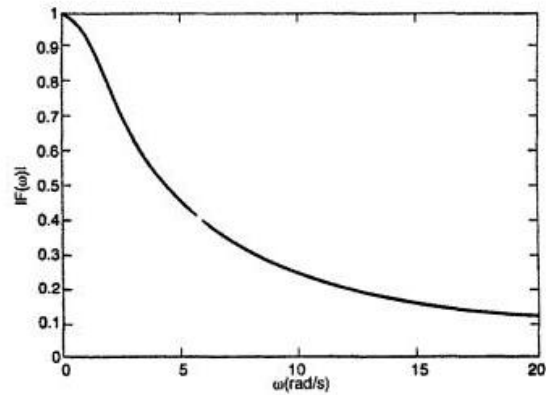


Figure 2.3 Magnitude spectrum of the signal with transform  $F(s) = 100/(s^2 + 40s + 100)$ .

When  $\zeta = 1$ , the poles  $p_1$  and  $p_2$  of  $F(s)$  are both equal to  $-\omega_n$ , and  $F(s)$  becomes:

$$F(s) = \frac{c}{(s+\omega_n)^2} \quad (2.76)$$

Taking the inverse transform gives

$$f(t) = cte^{-\omega_n t} \quad (2.77)$$

Since  $\omega_n$  is assumed to be strictly positive, when  $\zeta = 1$  both the poles are in the OLHP; in this case, the FT is

$$F(\omega) = \frac{c}{(j\omega + \omega_n)^2} \quad (2.78)$$

As  $\zeta$  varies from 1 to  $-1$ , the poles of  $F(s)$  trace out a circle in the complex plane with radius  $\omega_n$ . The loci of pole locations

is shown in Figure 2.4. Note that the poles begin at  $-\omega_n$  when  $\zeta = 1$ , then split apart and approach the  $j\omega$ -axis at  $\pm j\omega_n$  as  $\zeta \rightarrow 0$  and then move to  $\omega_n$  as  $\zeta \rightarrow -1$ . For  $-1 < \zeta < 1$ , the inverse transform of  $F(s)$  can be determined by first completing the square in the denominator of  $F(s)$ :

$$F(s) = \frac{c}{(s + \zeta\omega_n)^2 + \omega_d^2} \quad (2.79)$$

where

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} > 0 \quad (2.80)$$

Note that  $\omega_d$  is equal to the imaginary part of the pole  $p_1$  given by Equation 2.72. Using Table 2.4, we have that the inverse transform of  $F(s)$  is

$$f(t) = \frac{c}{\omega_d} e^{-\zeta\omega_n t} \sin \omega_d t \quad (2.81)$$

From Equation 2.81, it is seen that  $f(t)$  now contains a sinusoidal factor. When  $0 < \zeta < 1$ , the poles lie in the OLHP, and the signal is a decaying sinusoid. In this case, the FT is

$$F(\omega) = \frac{c}{(j\omega + \zeta\omega_n)^2 + \omega_d^2} \quad (2.82)$$

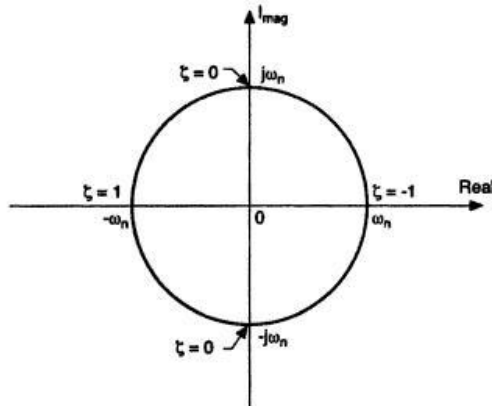


Figure 2.4 Loci of poles of  $F(s)$  as  $\zeta$  varies from 1 to  $-1$ .

The magnitude spectrum  $|F(\omega)|$  is plotted in Figure 2.5 for the values  $c = \omega_n^2 = 100$  and  $\zeta = 0.3, 0.5, 0.7$ . Note that for  $\zeta = 0.5$  and  $0.3$ , a peak appears in  $|F(\omega)|$ . This corresponds to the sinusoidal oscillation resulting from the  $\sin(\omega_d t)$  factor in  $f(t)$ . Also note that as  $\zeta$  is decreased from  $0.5$  to  $0.3$ , the peak increases in magnitude, which signifies a longer duration oscillation in the signal  $f(t)$ . This result is expected since the poles are approaching the  $j\omega$ -axis of the complex plane as  $\zeta \rightarrow 0$ . As  $\zeta \rightarrow 0$ , the peak in  $|F(\omega)|$  approaches  $\infty$ , so that  $|F(\omega)|$  does not exist (in the ordinary sense) in the limit as  $\zeta \rightarrow 0$ . When  $\zeta \rightarrow 0$ , the signal  $f(t)$  is purely oscillatory and does not have a FT in the ordinary sense. In addition, when  $\zeta < 0$  there is no FT (in any sense) since there is a pole of  $F(s)$  in the open right half-plane (ORHP).

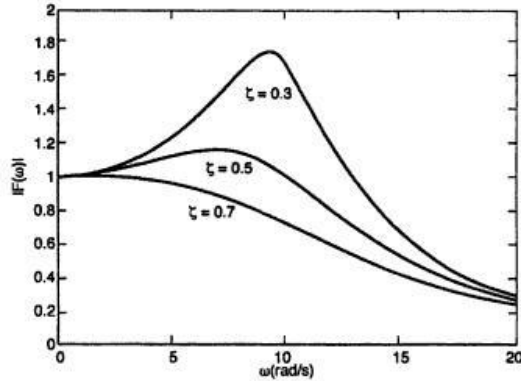


Figure 2.5 Magnitude spectrum of the signal with transform  $F(s) = 100/(s^2 + 20\zeta s + 100)$  and with  $\zeta = 0.7, 0.5, 0.3$ .

The above results lead to the following generalized properties of the magnitude spectrum  $|F(\omega)|$  of a signal  $f(t)$  whose Laplace transform  $F(s)$  is rational with all poles in the OLHP:

- If the poles of  $F(s)$  are real, the magnitude spectrum  $|F(\omega)|$  simply rolls off to zero as  $\omega \rightarrow \infty$ , starting with a peak value at  $\omega = 0$  of  $F(0)$ .
- If  $F(s)$  has a complex conjugate pair of poles at  $s = \sigma \pm j\omega_d$  and if the ratio  $\sigma/\omega_d$  is sufficiently small, then  $|F(\omega)|$  will have a peak located approximately at  $\omega = \omega_d$ .

### 2.3.2 Numerical Computation of the Fourier Transform

In many applications, the signal  $f(t)$  cannot be given in function form; rather, all one has are a set of sample values  $f(k) = f(kT)$ , where  $k$  ranges over a subset of integers and  $T$  is the sampling interval. Without loss of generality, we can assume that  $k$  starts with  $k = 0$ . Also, since all signals arising in practice are of finite duration in time, we can assume that  $f(k)$  is zero for all  $k \geq N$  for some positive integer  $N$ . The problem is then to determine the FT of  $f(t)$  using the sample values  $f(k) = f(kT)$  for  $k = 0, 1, 2, \dots, N-1$ .

One could also carry out a discrete-time analysis by taking the  $N$ -point DFT  $F_n$  of the sampled signal  $f(k)$ . In the previous section, it was shown that  $F_n$  is equal to  $F(\frac{2\pi n}{N})$  for  $n = 0, 1, 2, \dots, N-1$ , where  $F(\frac{2\pi n}{N})$  is the DTFT  $F(\Omega)$  of  $f(k)$  with the frequency variable  $\Omega$  evaluated at  $2\pi n/N$ . Hence the discrete-time counterpart of the frequency spectrum can be determined from  $F_n$ . For details on this, see "Further Reading."

Again letting  $F(\omega)$  denote the FT of  $f(t)$ , we can carry out a numerical computation of the FT as follows. First, since  $f(t)$  is zero for  $t < 0$  and  $t \geq NT$ , from the definition of the FT in Equation 2.1 we have

$$F(\omega) = \int_0^{NT} f(t) e^{-j\omega t} dt \quad (2.83)$$

Assuming that  $f(t)$  is approximately constant over each  $T$ -second interval  $[(k-1)T, kT]$ , we obtain the following approximation to Equation 2.83:

$$F(\omega) = \sum_{k=0}^{N-1} \left[ \int_{kT}^{kT+T} e^{-j\omega t} \right] f(k) \quad (2.84)$$

Then carrying out the integration in the right-hand side of Equation 2.84 gives

$$F(\omega) = \frac{1-e^{-j\omega T}}{j\omega} \sum_{k=0}^{N-1} e^{-j\omega kT} f(k) \quad (2.85)$$

Finally, setting  $\omega = 2\pi n/NT$  in Equation 2.85 yields

$$F\left(\frac{2\pi n}{NT}\right) = \frac{1-e^{-j2\pi n/N}}{j2\pi n/NT} F_n \quad (2.86)$$

where  $F_n$  is the DFT of  $f(k)$  given by Equation 2.46.

It should be stressed that the relationship in Equation 2.86 is only an approximation; that is, the right-hand side of Equation 2.86 is an approximation to  $F(2\pi n/NT)$ . In general, the approximation is more accurate the larger  $N$  is and/or the smaller  $T$  is. For a good result, it is also necessary that  $f(t)$  be suitably small for  $t < 0$  and  $t \geq NT$ .

As an example, let  $f(t)$  be the 2-second pulse given by  $f(t) = 1$  for  $0 \leq t \leq 2$  and  $f(t) = 0$  for all other  $t$ . Using the time shift property in Table 2.1 and the FT of a rectangular pulse given in Table 2.2, we have that the FT of  $f(t)$  is

$$F(\omega) = \frac{2}{\omega} \sin(\omega) e^{-j\omega} \quad (2.87)$$

A MATLAB program (adapted from [4]) for computing the exact magnitude spectrum  $|F(\omega)|$  and the approximation based on Equation 2.86 is given in Figure 2.6. The program was run for the case when  $N = 128$  and  $T = 0.1$ , with the plots shown in Figure 2.7. Note that the approximate values are fairly close, at least for frequencies over the span of the main lobe. A better approximation can be achieved by increasing  $N$  and/or decreasing  $T$ . The reader is invited to try this using the program in Figure 2.6.

```
N = input('Input N: ');
T = input('Input T: ');
t = 0:T:2;
f = [ones(1,length(t)) zeros(1,N-length(t))];
Fn = fft(f);
gam = 2*pi/N/T;
n = 0:10/gam;
Fapp = (1-exp(-j*n*gam*T))/j/n/gam*Fn;
w = 0:.05:10;
Fexact = 2*sin(w)./w;
plot(n*gam,abs(Fapp(1:length(n))), 'og', w,abs(Fexact), 'b')
```

Figure 2.6 MATLAB program for computing the Fourier transform of the 2-second pulse.

### 2.3.3 Solution of Differential Equations

One of the major applications of the Laplace transform is in solving linear differential equations. To pursue this, we begin by

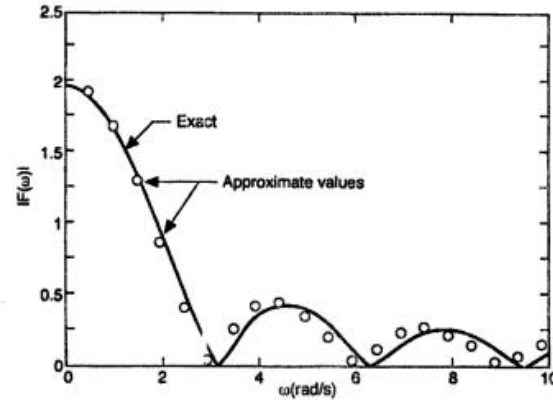


Figure 2.7 Exact and approximate magnitude spectra of the 2-second pulse.

considering the first-order linear constant-coefficient differential equation given by

$$\dot{f}(t) + af(t) = w(t), \quad t \geq 0 \quad (2.88)$$

where  $\dot{f}(t)$  is the derivative of  $f(t)$  and  $w(t)$  is an arbitrary real-valued function of  $t$ . To solve Equation 2.88, we apply the Laplace transform to both sides of the equation. Using linearity and the derivative properties of the Laplace transform given in Table 2.3, we have

$$sF(s) - f(0) + aF(s) = W(s) \quad (2.89)$$

where  $F(s)$  is the transform of  $f(t)$ ,  $W(s)$  is the transform of  $w(t)$ , and  $f(0)$  is the initial condition. Then solving Equation 2.89 for  $F(s)$  gives

$$F(s) = \frac{1}{s+a} [f(0) + W(s)] \quad (2.90)$$

Taking the inverse Laplace transform of  $F(s)$  then yields the solution  $f(t)$ . For example, if  $w(t)$  is the unit-step function  $u(t)$  and  $a \neq 0$ , then  $W(s) = 1/s$  and  $F(s)$  becomes

$$F(s) = \frac{1}{s+a} \left[ f(0) - \frac{1}{s} \right] = \frac{f(0)s+1}{(s+a)s} = \frac{f(0)-\frac{1}{a}}{s+a} + \frac{1}{s} \quad (2.91)$$

Taking the inverse transform gives

$$f(t) = \left[ f(0) - \frac{1}{a} \right] e^{-at} + \frac{1}{a}, \quad t \geq 0 \quad (2.92)$$

Now consider the second-order differential equation

$$\ddot{f}(t) + a_1 \dot{f}(t) + a_0 f(t) = w(t) \quad (2.93)$$

Again using the derivative property of the Laplace transform, taking the transform of both sides of Equation 2.93 we obtain

$$s^2 F(s) - sf(0) - \dot{f}(0) + a_1 [sF(s) - f(0)] - a_0 F(s) = W(s) \quad (2.94)$$

where  $f(0)$  and  $\dot{f}(0)$  are the initial conditions. Solving Equation 2.94 for  $F(s)$  yields

$$F(s) = \frac{1}{s^2 + a_1 s + a_0} [s^2 f(0) + \dot{f}(0) + a_1 f(0) + W(s)] \quad (2.95)$$

For example, if  $a_0 = 2$ ,  $a_1 = 3$ , and  $w(t) = u(t)$ , then

$$\begin{aligned} F(s) &= \frac{1}{(s+1)(s+2)} \left[ f(0)s + \dot{f}(0) + 3f(0) + \frac{1}{s} \right] \\ F(s) &= \frac{f(0)s^2 + [\dot{f}(0) + 3f(0)]s + 1}{(s+1)(s+2)s} \\ F(s) &= \frac{2f(0) + \dot{f}(0) - 1}{s+1} + \frac{-\frac{5}{2}f(0) - \dot{f}(0) + \frac{1}{2}}{s+2} \\ &\quad + \frac{\frac{1}{2}}{s} \end{aligned} \quad (2.96)$$

Inverse transforming Equation 2.96 gives

$$\begin{aligned} f(t) &= [2f(0) + \dot{f}(0) - 1]e^{-t} \\ &\quad + [-2.5f(0) - \dot{f}(0) + 0.5]e^{-2t} + 0.5, \quad t \geq 0 \end{aligned} \quad (2.97)$$

For the general case, consider the  $n$ th-order linear constant-coefficient differential equation

$$f^{(n)}(t) + \sum_{i=0}^{n-1} a_i f^{(i)}(t) = w(t) \quad (2.98)$$

where  $f^{(i)}(t)$  is the  $i$ th derivative of  $f(t)$ . Given  $w(t)$  and the initial conditions  $f(0), \dot{f}(0), \dots, f^{(n-1)}(0)$ , the solution  $f(t)$  to Equation 2.98 is unique. The solution can be determined by taking the transform of both sides of Equation 2.98 and solving for  $F(s)$ . This yields

$$F(s) = \frac{1}{D(s)} [N(s) + W(s)] \quad (2.99)$$

where  $D(s)$  is the polynomial

$$D(s) = s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0 \quad (2.100)$$

and  $N(s)$  is a polynomial in  $s$  of the form

$$N(s) = b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_1s + b_0 \quad (2.101)$$

The coefficients  $b_0, b_1, \dots, b_{n-1}$  of  $N(s)$  depend on the values of the  $n$  initial conditions  $f(0), \dot{f}(0), \dots, f^{(n-1)}(0)$ . The relationship between the  $b_i$  and the initial conditions is given by the matrix equation

$$b = Px \quad (2.102)$$

where  $b$  and  $x$  are the column vectors

$$b = \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_{n-2} \\ b_{n-1} \end{bmatrix} \quad x = \begin{bmatrix} f(0) \\ \dot{f}(0) \\ \vdots \\ f^{(n-2)}(0) \\ f^{(n-1)}(0) \end{bmatrix} \quad (2.103)$$

and  $P$  is the  $n$ -by- $n$  matrix given by

$$P = \begin{bmatrix} a_1 & a_2 & \dots & a_{n-2} & a_{n-1} & 1 \\ a_2 & a_3 & \dots & a_{n-1} & 1 & 0 \\ \vdots & \vdots & & \vdots & \vdots & \vdots \\ a_{n-1} & 1 & \dots & 0 & 0 & 0 \\ 1 & 0 & \dots & 0 & 0 & 0 \end{bmatrix} \quad (2.104)$$

The matrix  $P$  given by Equation 2.104 is invertible for any values of the  $a_i$ , and thus there is a one-to-one and onto correspondence between the set of initial conditions and the coefficients of the polynomial  $N(s)$  in Equation 2.101. In particular, this implies that for any given vector  $b$  of coefficients of  $N(s)$ , there is a vector  $x$  of initial conditions that results in the polynomial  $N(s)$  with the given coefficients. From Equation 2.102, it is seen that  $x = P^{-1}b$  where  $P^{-1}$  is the inverse of  $P$ .

Once  $N(s)$  is computed using Equation 2.102, the solution  $f(t)$  to Equation 2.98 can then be determined by inverse transforming Equation 2.99. If  $W(s)$  is a rational function of  $s$ , then the right-hand side of Equation 2.99 is rational in  $s$  and thus, in this case,  $f(t)$  can be computed via a partial fraction expansion.

An interesting consequence of the above constructions is the following characterization of a real-valued function  $f(t)$  whose Laplace transform  $F(s)$  is rational. Suppose that

$$F(s) = \frac{N(s)}{D(s)} \quad (2.105)$$

where  $D(s)$  and  $N(s)$  are given by Equations 2.100 and 2.101, respectively. Then comparing Equations 2.105 and 2.99 shows that  $f(t)$  is the solution to the  $n$ th-order homogeneous equation

$$f^{(n)}(t) + \sum_{i=0}^{n-1} a_i f^{(i)}(t) = 0 \quad (2.106)$$

with the initial conditions given by  $x = P^{-1}b$ , where  $x$  and  $b$  are defined by Equation 2.103. Hence, any function  $f(t)$  having a rational Laplace transform is the solution to a homogeneous differential equation. This result is of fundamental importance in the theory of systems and controls.

### 2.3.4 Solution of Difference Equations

The discrete-time counterpart to the solution of differential equations using the Laplace transform is the solution of difference equations using the  $z$ -transform. We begin by considering the first-order linear constant-coefficient difference equation

$$f(k+1) + af(k) = w(k), \quad k \geq 0 \quad (2.107)$$

where  $w(k)$  is an arbitrary real-valued function of the integer variable  $k$ . Taking the  $z$ -transform of Equation 2.107 using the linearity and left shift properties given in Table 2.6 yields

$$zF(z) - f(0)z + aF(z) = W(z) \quad (2.108)$$

where  $F(z)$  is the  $z$ -transform of  $f(k)$  and  $f(0)$  is the initial condition. Then solving Equation 2.108 for  $F(z)$  gives

$$F(z) = \frac{1}{z+a} [f(0)z + W(z)] \quad (2.109)$$

For example, if  $w(k)$  is the unit-step function  $u(k)$  and  $a \neq 1$ , then  $W(z) = z/(z-1)$  and  $F(z)$  becomes

$$\begin{aligned} F(z) &= \frac{1}{z+a} \left[ f(0)z + \frac{z}{z-1} \right] \\ F(z) &= \frac{f(0)z(z-1) + z}{(z+a)(z-1)} \end{aligned} \quad (2.110)$$

**Proper rational function:** A rational function  $N(s)/D(s)$  where the degree of  $N(s)$  is less than or equal to the degree of  $D(s)$ .

**Rational function:** A ratio of two polynomials  $N(s)/D(s)$  where  $s$  is a complex variable.

**Region of convergence:** The set of all complex numbers for which a transform exists (i.e., is well defined) in the ordinary sense.

**Residues:** The values of the numerator constants in a partial fraction expansion of a rational function.

**Strictly proper rational function:** A rational function  $N(s)/D(s)$  where the degree of  $N(s)$  is strictly less than the degree of  $D(s)$ .

**Two-sided (or bilateral) transform:** A transform that operates on a function  $f(t)$  defined for  $-\infty < t < \infty$ .

**Zeros of a rational function  $N(s)/D(s)$ :** The values of  $s$  for which  $N(s) = 0$ , assuming that  $N(s)$  and  $D(s)$  have no common factors.

For a thorough development of the z-transform, see Jury, E.I. 1964. *Theory and Application of the z-Transform Method*, Wiley, New York.

Treatments of the Fourier, Laplace, and z-transforms can be found in textbooks on signals and systems, such as

Oppenheim, A.V. and Willsky, A.S., 1983. *Signals and Systems*, Prentice Hall, Englewood Cliffs, NJ.

Ziemer, R.E., Tranter, W.H. and Fannin, D.R., 1993. 3rd ed., Macmillan, New York.

For a development that is integrated together with the MATLAB software package, see [4].

## References

- [1] Barker, R.H., The pulse transfer function and its applications to sampling servo systems, *Proc. IEEE*, 99, Part IV, 302–317, 1952.
- [2] Fourier, J.B.J., *The Analytical Theory of Heat*, Cambridge, (transl. A. Freeman) 1878.
- [3] Jury, E.I., Analysis and synthesis of sampled-data control systems, *Communications and Electronics*, 1954, 1–15.
- [4] Kamen, E.W. and Heck, B.S., *Fundamentals of Signals and Systems with MATLAB*, Prentice Hall, Englewood Cliffs, NJ, 1996.
- [5] Ragazzini, J.R. and Zadeh, L. A., The analysis of sampled-data systems, *Trans. AIEE*, 71, Part II:225–232, 1952.

## Further Reading

A mathematically rigorous development of the FT can be found in Papoulis, A. 1962. *The Fourier Integral and Its Applications*, McGraw-Hill, New York.

Bracewell, R.M. 1965. *The Fourier Transform and Its Applications*, McGraw-Hill, New York.

An in-depth treatment on the use of the Fast Fourier transform in Fourier analysis can be found in

Brigham, E.O. 1988. *The Fast Fourier Transform and Its Applications*, Prentice Hall, Englewood Cliffs, NJ.

For a detailed development of the Laplace transform, see Rainville, E.D. 1963. *The Laplace Transform: An Introduction*, Macmillan, New York.